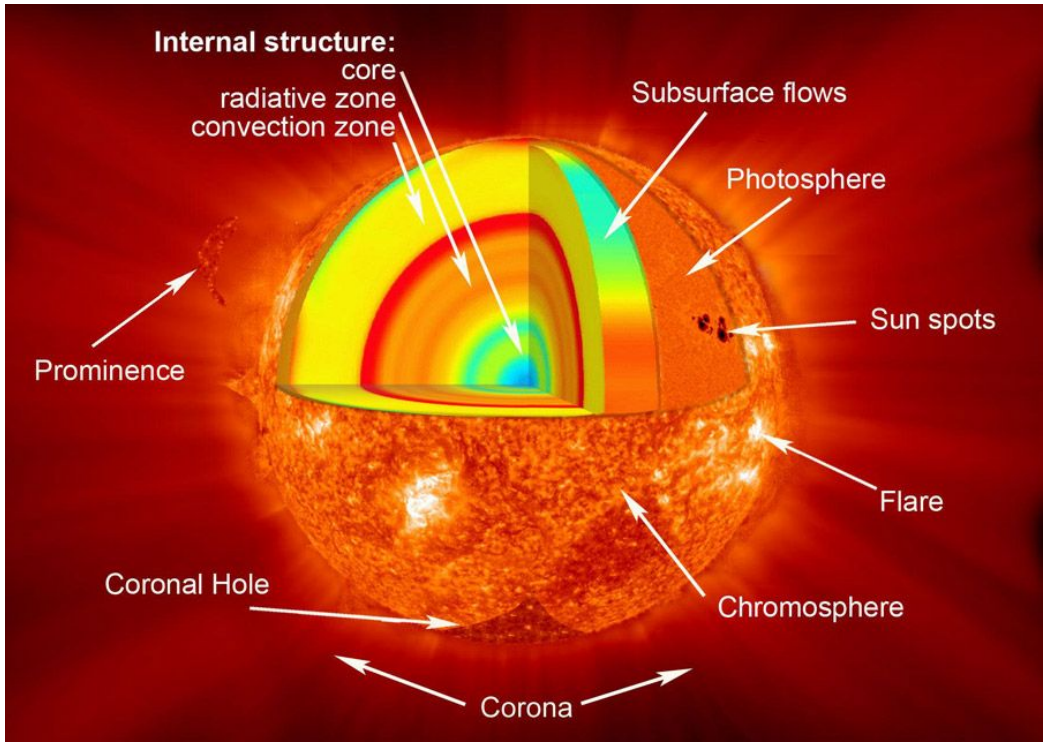
Interactions Between the Earth, Moon, & Sun

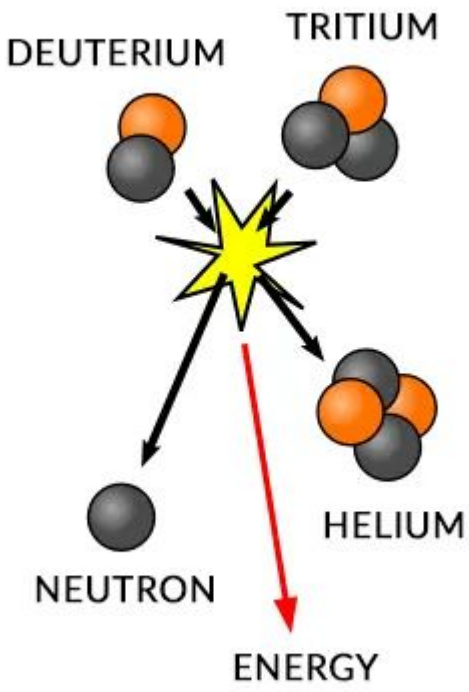
The Sun

- The Sun was born approximately **5 billion** years ago
- It is responsible for our **climate** and provides all of the **energy** for life on Earth



The Sun

- Inside the core of the Sun is where **nuclear fusion** occurs
- Two atoms of **hydrogen** fuse together to form one atom of **helium**
- This reaction releases tremendous amounts of **energy**



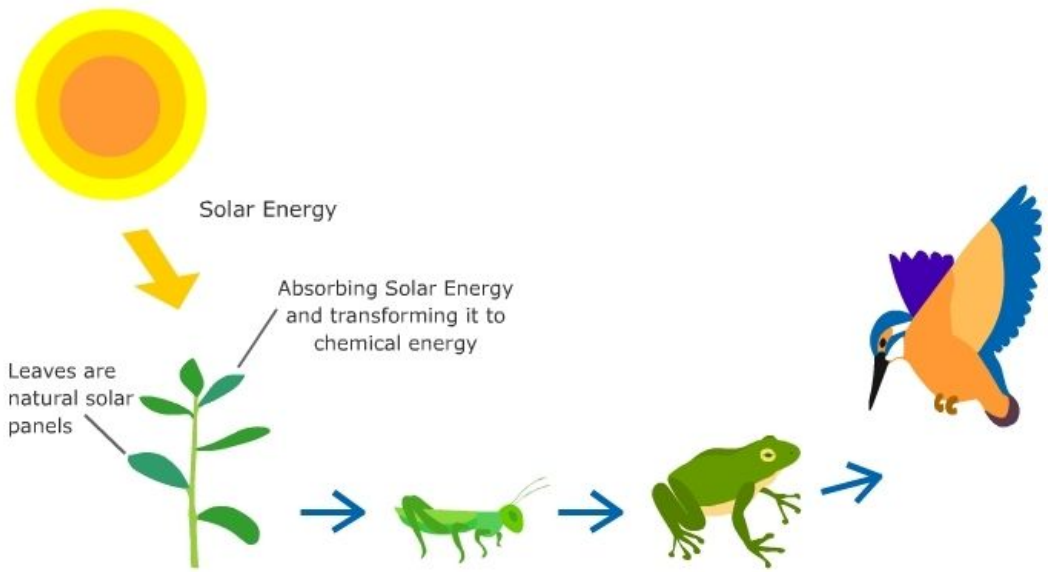
How much energy?

- During fusion, each **gram** of hydrogen releases 90 000 **million** kJ of energy
- That's as much energy as you would get if you ate **1 trillion** pizza slices!
- And that's only for **1 g** of hydrogen – the Sun has **2.0 x 10^{23} g** of hydrogen!

Nuclear Fusion Breakthrough - Dec 2022

What is this energy used for?

- Plants convert sunlight into **chemical** energy
- Animals **eat** plants that are grown in sunlight to obtain **chemical** energy
- Solar panels convert sunlight into **electrical** energy



Plant is eaten by grasshoper is eaten by frog is eaten by bird.
Stored chemical energy is transferred from the plant to the grasshopper, to the frog, to the bird, enabling each in turn to function as a living organism.

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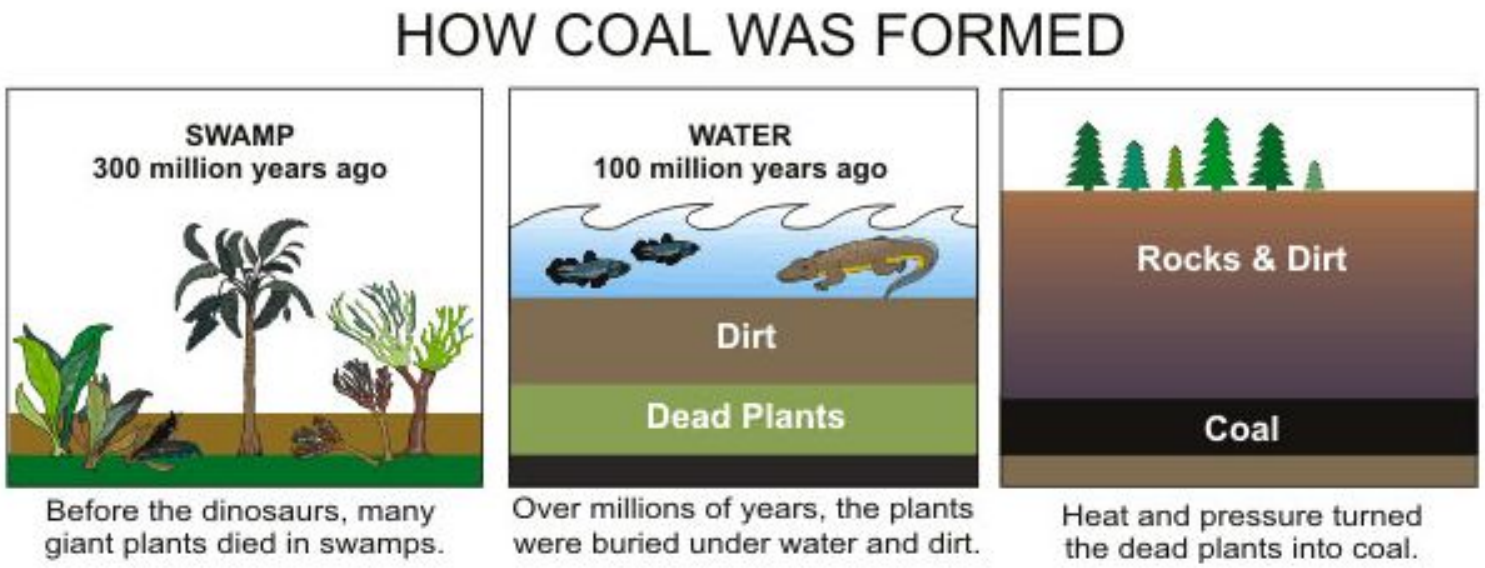
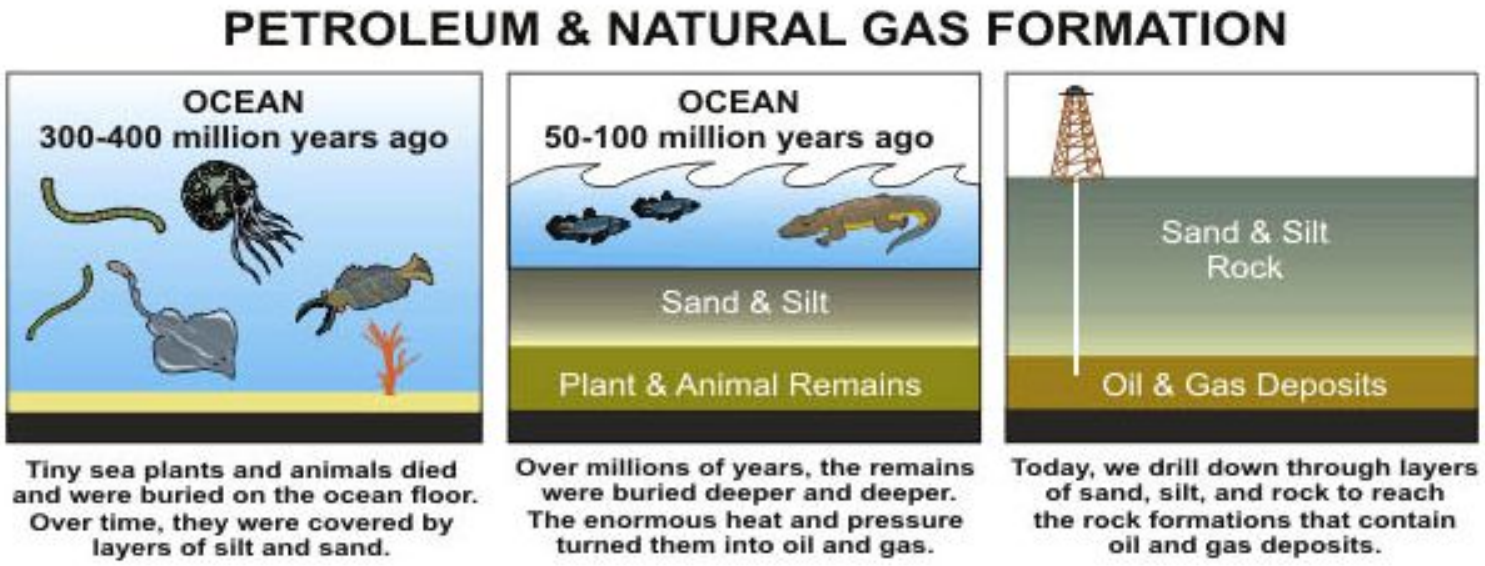
Fossil Fuels

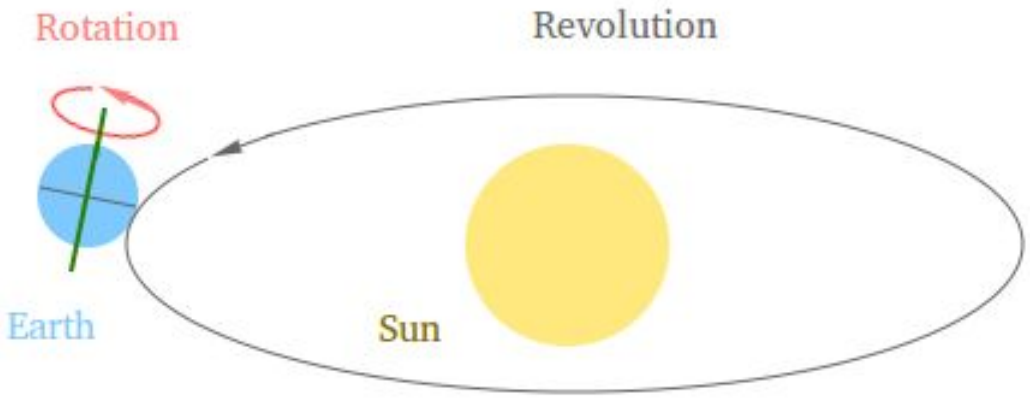
Stored Energy from the Sun





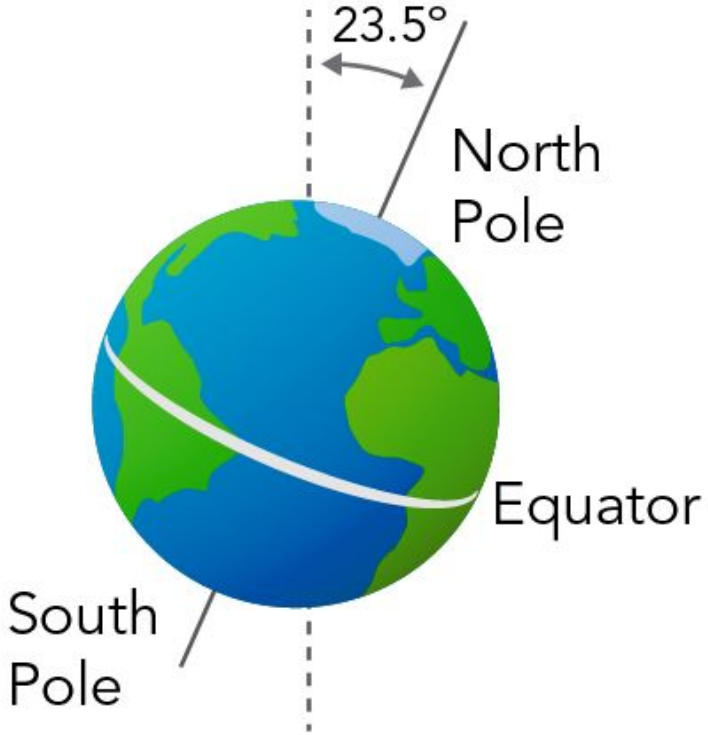
Rotation vs. Revolution

- Rotation is the spinning of an object on its **axis**
- Revolution is the movement of one object travelling **around** another object
- Earth's revolution around the Sun is **elliptical** in shape



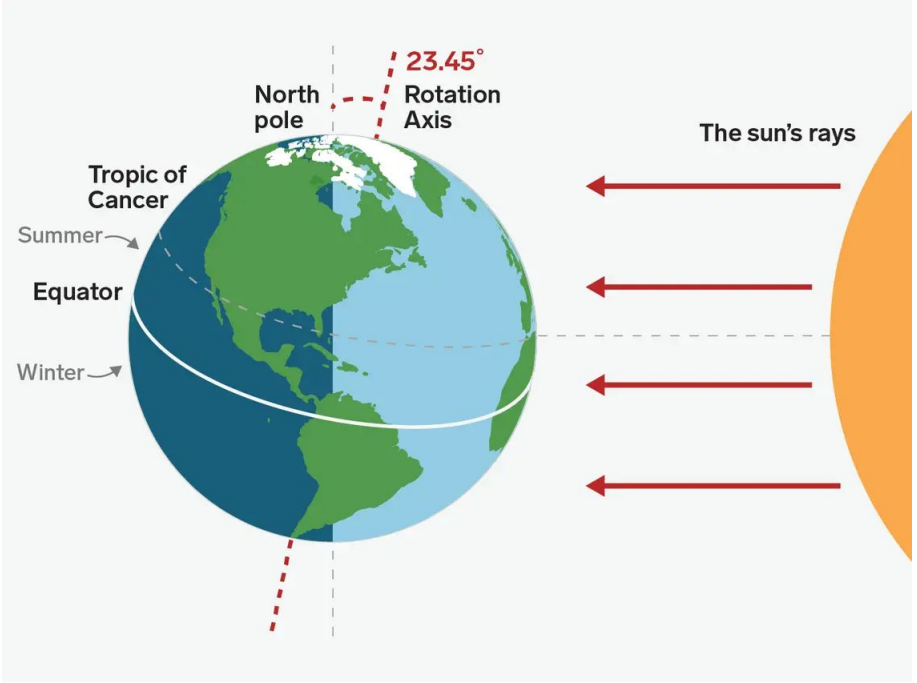
Earth's Tilt

- Earth's axis is tilted **23.5°** from **vertical** when compared to its orbit around the Sun
- This causes each of Earth's **hemispheres** to point towards or away from the Sun during its **revolution** around the Sun



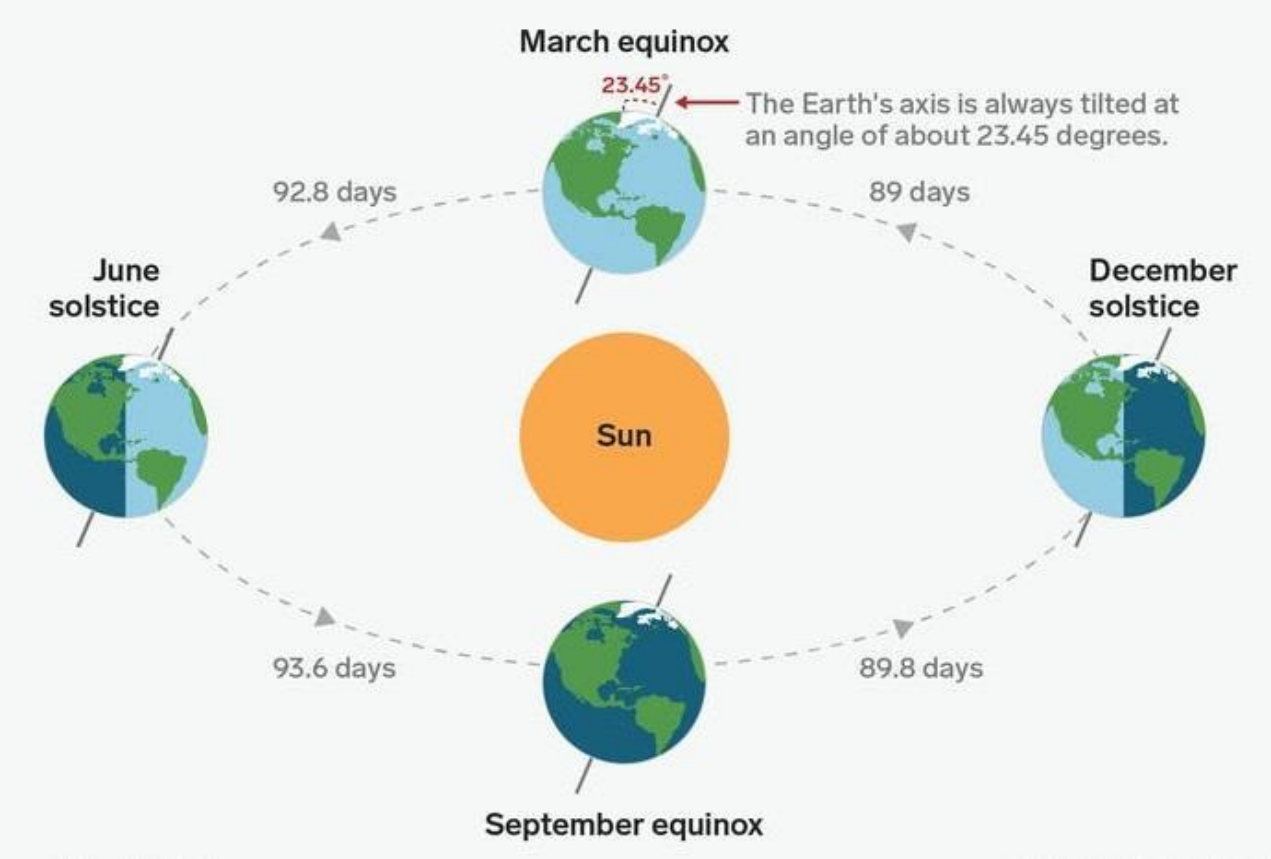
The Reason for Seasons

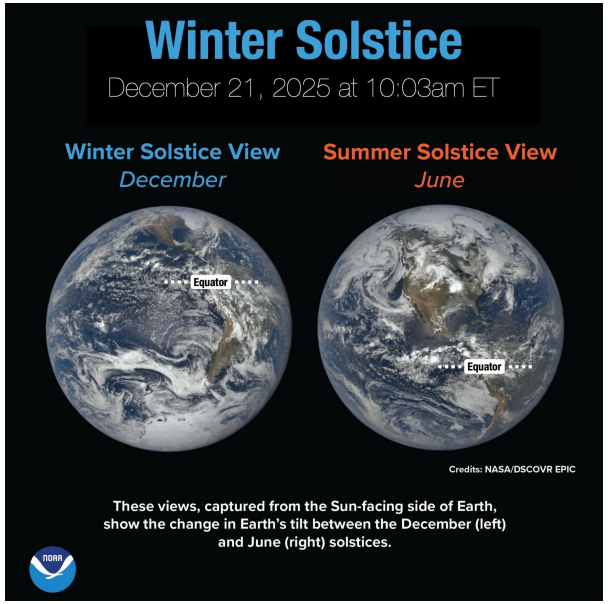
- When the Northern hemisphere is pointed **towards** the Sun, we get more **direct** sunlight, giving us **summer**
- When the Northern hemisphere is pointed **away** from the Sun, we get **indirect** sunlight, giving us **winter**



WHY DO
WE HAVE
SEASONS?
SciShow

Solstice & Equinox



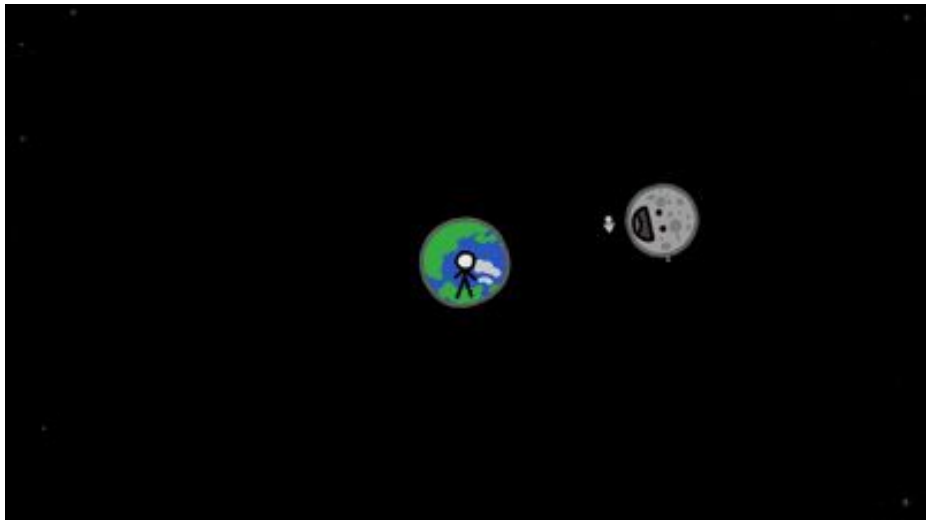


The Moon

- The Moon in **non-luminous** – it doesn't produce it's own light
- We see the Moon (and other planets) as it **reflects** the light from the Sun
- However, the **illuminated** side of the Moon doesn't always face the Earth

Synchronous Rotation

- The Moon rotates at the **same** speed that it revolves
- This means that it spins on its axis at a speed that **matches** how fast it's going around the Earth
- This is what causes the Moon to always have the **same** side facing us



The Lunar Cycle

- When sunlight reflects off the Moon's **far** side (the side we can't see) it is called a **New** Moon
- When sunlight reflects off the **near** side, we call it a **Full** Moon

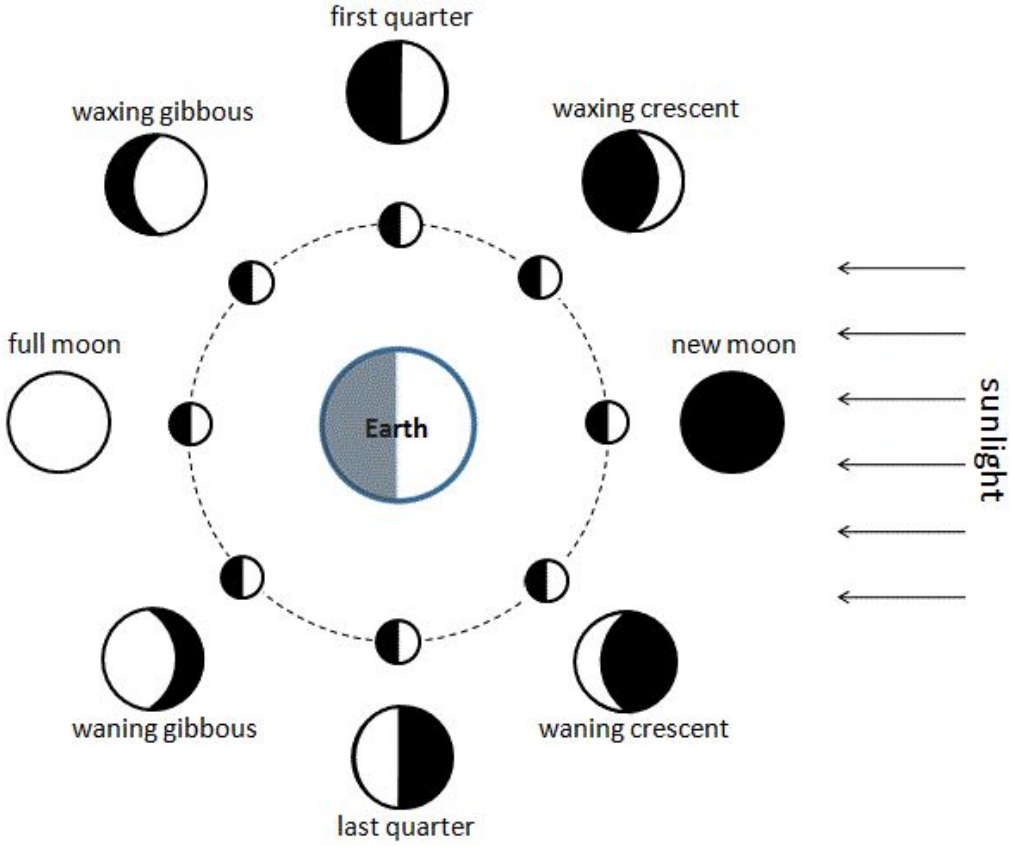


Image of a total lunar eclipse from the Moon's surface captured by Firefly's Blue Ghost Mission 1 lunar lander on March 14, 2025. The image shows the sun about to emerge from totality behind Earth.

Firefly Aerospace

Indigenous Moon

From the full Moon cycles, it was noted that certain things happened in the environment. The weather, plants, animals and temperature seemed to follow the cycles of the Moon. One of the patterns that became apparent was that there were 13 full moons that occurred before everything seemed to start all over again. Thus a 13 Moon cycle was identified.

When Europeans made contact with the First Nations peoples of the Americas, it was evident that some Indigenous peoples followed a lunar calendar depicted on the back of a turtle's shell. Turtle shells have 28 smaller outer edge scutes, representing the number of days from one full Moon to the next, and 13 larger central scutes, representing the 13 moon cycles. The turtle (Mikinak Ministik) held a special place in the various cultures of Indigenous people. One of the origins of the term Turtle Island is the lunar calendar.

Your turn

- Eclipses and the Northern Lights